

A two-stage stochastic programming approach incorporating spatially-explicit fire scenarios for optimal firebreak placement

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Interesting paper dealing with problem of firebreak location, including uncertainty on the performance via fire scenarios. The paper proposes two measures for this performance, the expected value and the CvaR, and a linear combination of both. A MIP model is proposed to obtain the solution, and computational experiments are carried out comparing different values for the parameters included in the model.

The model is well organized and well written. The MIP model is a new proposal for this problem. The results regarding the performance are good results, although they lose effectivity when being tested with other fire simulations, which is a little bit disappointed.

Some minor changes and suggestions are shown in the following.

- 1) It is not clear the advantage of including the computational experiments Sub40 and Sub100. Sub20 is a 400ha landscape, of 20x20 cells, each of them being 100x100 m2. But Sub40 and Sub 100 are not clear. It is written that cells are 100x100 m2 also for these two cases, and Sub40 looks to be 40x40 cells, and Sub100 100x100 cells. However, in these two cases Figures 10 and 11 shows 1km to be the length of each side of the square.

Moreover, in section 4.3, the explanation about the MIP gap is not clear. It is normal the results for Sub40, increasing the MIP gap when increasing the parameter α , since there are more feasible combinations. But it is said later that for Sub60 (please note that Sub60 has not been included in the computational study) and Sub100 the MIP gap decreases when increasing α . It should be explained, if I have well understood, that Sub 100 is a 100x100 cells case (10,000 cells), 1% is 100 cells and 5% is 500, and the number of scenarios is 100. In such a case, the solution can be put firebreaks around the ignition point of the 100 scenarios (but these are 1% of the possible scenarios). There are more possible combinations, but the solution is easier to be reached. The subsequent explanation is not clear, and there is a number 8 missing in the formula of page 14 ($\alpha \text{ card}(V) = 8 \text{ card}(\Omega)$). After this explanation, it could be shown the performance with 1000 scenarios, as it has been done with Sub20.

In summary, these cases must be better explained, and perhaps, their study extended a little bit. And one of the conclusions should be that the number of scenarios should be big enough related to the number of firebreaks to be located.

- 2) One minor question for figures 1 and 2 is that it is not well understood that some cells burn since the legend put a red dot into a square with black line, and it is clear when the background is dark, but it is not so clear when the background is light green. The border of the square is missing for light green cells burning, and it is not clear when seeing the figure.

Also, in Figure 2 would be nice to see the saved cells.

- 3) Page 6, fifth line from the end, a small mistake: the subindex of w is v instead of ω .

- 4) Page 7, explanation of the constraints:
 - a. *In particular, the constraint (8f) force...* It should be constraint (8e), as well as for being nonnegative is constraint (8f) instead of (8g).
 - b. From equation (8d) it looks that all the adjacent cells burn or are firebreaks. But, it is only for those cells belonging to $\delta^+_{\omega}(v)$, is it mean that this constraint applies only for those cells reached from v in the scenario simulated without firebreaks? It should be clearly explained.
- 5) How are defined the scenarios that you need 5 simulations for each number of scenarios? It means that, for instance, the number of scenarios is 100, and the simulation is repeated five times with 100 scenarios, isn't it? Why is not defined directly with 500 scenarios? Or are obtained different solutions from the model in each of the five repetitions?
- 6) Page 9, second paragraph of 4.2: the following sentence is not finished: *These graphs are incorporated into the mathematical formulation by the.*
- 7) Section 4: please, put the figures where are named in the text, it has been difficult to pass pages and pages following the referenced figures. As well as put the caption of each image closer to the image, not closer to the next image.
- 8) Finally, Figure 9 is very disappointing. Random location achieves similar solutions to the others with 1% and 2% of firebreaks, and similar with other percentages using only 20 scenarios. And, regarding the images, it looks more important using a number big enough of scenarios, than the parameter λ . It is a key analysis; it is the validation of the model. An image with the different values of λ would be welcomed to compare the differences (one image or three images, one per each number of scenarios).